

Customer Retention Intelligence - Statistical Analysis

Abhinav Sriram

Overview

This analysis evaluated relationships identified during exploratory data analysis and determines which customer characteristics remain associated with churn after accounting for other factors.

This analysis focuses on: -Contract type -Customer tenure -Monthly charges -Premium technical support -Internet type -Customer referrals -Demographic characteristics

```
library(DBI)
library(RSQLite)
library(dplyr)
library(broom)
library(here)

knitr::opts_chunk$set(
  echo = TRUE,
  message = FALSE,
  warning = FALSE
)
```

#Load Data from SQLite

The analytical data is retrieved directly from the SQLite database created during the data-engineering phase.

```
db_path = here(
  "database", "customer_retention.db"
)

con <- dbConnect(SQLite(), db_path)
```

```
query <- "
SELECT
  c.customer_id,
  c.age,
  c.married,
  c.number_of_dependents,
  c.number_of_referrals,
  c.tenure_months,
  c.churn,

  a.contract,
  a.payment_method,
```

```

    a.monthly_charge,

    s.internet_service,
    s.internet_type,
    s.premium_tech_support

FROM customers AS c

JOIN customer_accounts AS A
  ON c.customer_id = a.customer_id

JOIN customer_services AS s
  ON c.customer_id = s.customer_id

WHERE c.churn IS NOT NULL;"

customer_data <- dbGetQuery(con, query)

dbDisconnect(con)

```

```
#Initial Inspection
```

```
class(customer_data)
```

```
## [1] "data.frame"
```

```
dim(customer_data)
```

```
## [1] 6589 13
```

```
head(customer_data)
```

```
##  customer_id age married number_of_dependents number_of_referrals
## 1  0002-ORFBO  37    Yes                      0                  2
## 2  0003-MKNFE  46    No                       0                  0
## 3  0004-TLHLJ  50    No                       0                  0
## 4  0011-IGKFF  78    Yes                      0                  1
## 5  0013-EXCHZ  75    Yes                      0                  3
## 6  0013-MHZWF  23    No                       3                  0
##  tenure_months churn      contract  payment_method monthly_charge
## 1              9      0      One Year   Credit Card      65.6
## 2              9      0 Month-to-Month Credit Card      NA
## 3              4      1 Month-to-Month Bank Withdrawal  73.9
## 4             13      1 Month-to-Month Bank Withdrawal  98.0
## 5              3      1 Month-to-Month Credit Card      83.9
## 6              9      0 Month-to-Month Credit Card      69.4
##  internet_service internet_type premium_tech_support
## 1              Yes      Cable      Yes
## 2              Yes      Cable      No
## 3              Yes  Fiber Optic      No
## 4              Yes  Fiber Optic      No
## 5              Yes  Fiber Optic      Yes
## 6              Yes      Cable      Yes
```

The analytical dataset should contain 6,589 customers representing customers who either stayed or churned. Newly joined customers are excluded because their long-term churn outcome has not yet been observed.

#Data Preparation

Categorical variables are converted to factors before statistical testing.

```
customer_data <- customer_data %>%
  mutate(churn = factor(churn, levels = c(0,1)),
         contract = factor(contract),
         premium_tech_support = factor(premium_tech_support),
         internet_type = factor(internet_type),
         internet_service = factor(internet_service),
         married = factor(married),
         payment_method = factor(payment_method)
  )
```

#Contract Type and Churn

EDA suggested that month-to-month customers experience substantially higher churn than customers with longer-term contracts.

The hypotheses are: **H0**: Contract type and churn are independent. **HA**: Contract type and churn are associated.

```
contract_table <- table(
  customer_data$contract,
  customer_data$churn
)
```

```
contract_table
```

```
##
##           0      1
## Month-to-Month 1547 1655
## One Year       1360  166
## Two Year       1813   48
```

```
contract_test <- chisq.test(contract_table)
```

```
contract_test
```

```
##
## Pearson's Chi-squared test
##
## data:  contract_table
## X-squared = 1695.7, df = 2, p-value < 2.2e-16
```

#Premium Technical Support and Churn

EDA showed substantially higher churn among customers without premium technical support.

The hypotheses are: **H0**: Premium technical support and churn are independent. **HA**: Premium technical support and churn are associated.

```
support_table <- table(customer_data$premium_tech_support, customer_data$churn)
```

```
support_table
```

```
##
##              0      1
##   No          1802 1446
##   Not Applicable 1231  113
##   Yes          1687  310
```

```
support_test <- chisq.test(support_table)
```

```
support_test
```

```
##
##   Pearson's Chi-squared test
##
## data:  support_table
## X-squared = 842.68, df = 2, p-value < 2.2e-16
```

#Internet Type and Churn

EDA showed differences in churn rates among fiber optic, cable, DSL, and customers without internet service.

The hypotheses are: **H0**: Internet type and churn are independent. **HA**: Internet type and churn are associated.

```
internet_table <- table(
  customer_data$internet_type,
  customer_data$churn
)
```

```
internet_table
```

```
##
##              0      1
##   Cable          561  213
##   DSL            1230  307
##   Fiber Optic    1698 1236
##   Not Applicable 1231  113
```

```
internet_test <- chisq.test(internet_table)
```

```
internet_test
```

```
##
##   Pearson's Chi-squared test
##
## data:  internet_table
## X-squared = 590.44, df = 3, p-value < 2.2e-16
```

#Prepare Data for Logistic Regression The invalid negative monthly charges identified during data cleaning were converted to missing values.

For this statistical model, observations with missing monthly charges are excluded

```
sum(is.na(customer_data$monthly_charge))
```

```
## [1] 114
```

```
model_df <- customer_data %>%  
  filter(!is.na(monthly_charge))  
  
dim(model_df)
```

```
## [1] 6475 13
```

Create a numeric binary churn outcome for logistic regression.

```
model_df$churn_numeric <- as.integer(  
  as.character(model_df$churn)  
)
```

#Multivariable Logistic Regression

A logistic regression model is used to evaluate which variables remain associated with churn after accounting for other predictors.

```
churn_model <- glm(churn_numeric ~  
  contract +  
  tenure_months +  
  monthly_charge +  
  premium_tech_support +  
  internet_type +  
  number_of_referrals +  
  age +  
  number_of_dependents +  
  married,  
  data = model_df,  
  family = binomial)  
  
summary(churn_model)
```

```
##  
## Call:  
## glm(formula = churn_numeric ~ contract + tenure_months + monthly_charge +  
##   premium_tech_support + internet_type + number_of_referrals +  
##   age + number_of_dependents + married, family = binomial,  
##   data = model_df)  
##  
## Coefficients: (1 not defined because of singularities)  
##  
##               Estimate Std. Error z value Pr(>|z|)  
## (Intercept)    -0.748329   0.229602  -3.259 0.001117 **  
## contractOne Year    -1.345943   0.110105 -12.224 < 2e-16 ***  
## contractTwo Year    -2.380669   0.169780 -14.022 < 2e-16 ***  
## tenure_months     -0.039734   0.002446 -16.242 < 2e-16 ***  
## monthly_charge     0.021166   0.003458   6.122 9.27e-10 ***
```

```
## premium_tech_supportNot Applicable -0.687465 0.194451 -3.535 0.000407 ***
## premium_tech_supportYes -0.522343 0.098078 -5.326 1.00e-07 ***
## internet_typeDSL -0.356862 0.132417 -2.695 0.007039 **
## internet_typeFiber Optic -0.185202 0.162086 -1.143 0.253200
## internet_typeNot Applicable NA NA NA NA
## number_of_referrals -0.533354 0.034280 -15.559 < 2e-16 ***
## age 0.012916 0.002251 5.738 9.61e-09 ***
## number_of_dependents -0.674560 0.062243 -10.837 < 2e-16 ***
## marriedYes 1.618500 0.112890 14.337 < 2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 7727.3 on 6474 degrees of freedom
## Residual deviance: 4490.8 on 6462 degrees of freedom
## AIC: 4516.8
##
## Number of Fisher Scoring iterations: 7
```

#Odds Ratios

Logistic regression coefficients are exponentiated to produce odds ratios.

An odds ratio: -Greater than 1 indicate higher churn odds. -Less than 1 indicated lower churn odds. -Near 1 indicates little change in churn odds.

```
odds_ratios <- tidy(churn_model, exponentiate = TRUE, coef.int = TRUE)

odds_ratios
```

```
## # A tibble: 14 x 5
##   term                estimate std.error statistic  p.value
##   <chr>              <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)        0.473    0.230     -3.26  1.12e- 3
## 2 contractOne Year    0.260    0.110    -12.2  2.31e-34
## 3 contractTwo Year    0.0925   0.170    -14.0  1.14e-44
## 4 tenure_months      0.961    0.00245  -16.2  2.53e-59
## 5 monthly_charge     1.02     0.00346   6.12  9.27e-10
## 6 premium_tech_supportNot Applicable 0.503    0.194    -3.54  4.07e- 4
## 7 premium_tech_supportYes 0.593    0.0981   -5.33  1.00e- 7
## 8 internet_typeDSL    0.700    0.132    -2.69  7.04e- 3
## 9 internet_typeFiber Optic 0.831    0.162    -1.14  2.53e- 1
## 10 internet_typeNot Applicable NA      NA      NA      NA
## 11 number_of_referrals 0.587    0.0343  -15.6  1.38e-54
## 12 age               1.01     0.00225   5.74  9.61e- 9
## 13 number_of_dependents 0.509    0.0622  -10.8  2.29e-27
## 14 marriedYes        5.05     0.113    14.3  1.29e-46
```

#Statistically Significant Predictors

The following table displays predictors with p-values below 0.05.

```
significant_predictors <- odds_ratios %>%
  filter(
    term != "(Intercept)",
    p.value < 0.05
  ) %>%
  arrange(p.value)

significant_predictors
```

```
## # A tibble: 11 x 5
##   term                                estimate std.error statistic  p.value
##   <chr>                                <dbl>     <dbl>     <dbl>    <dbl>
## 1 tenure_months                      0.961     0.00245    -16.2 2.53e-59
## 2 number_of_referrals                 0.587     0.0343    -15.6 1.38e-54
## 3 marriedYes                        5.05      0.113      14.3 1.29e-46
## 4 contractTwo Year                   0.0925    0.170     -14.0 1.14e-44
## 5 contractOne Year                   0.260     0.110     -12.2 2.31e-34
## 6 number_of_dependents               0.509     0.0622    -10.8 2.29e-27
## 7 monthly_charge                    1.02      0.00346     6.12 9.27e-10
## 8 age                               1.01      0.00225     5.74 9.61e- 9
## 9 premium_tech_supportYes           0.593     0.0981     -5.33 1.00e- 7
## 10 premium_tech_supportNot Applicable 0.503     0.194     -3.54 4.07e- 4
## 11 internet_typeDSL                 0.700     0.132     -2.69 7.04e- 3
```

#Save Results

The logistic regression results are exported for use in later project stages and documentation.

```
write.csv(
  odds_ratios,
  here(
    "data", "processed", "logistic_refression_results.csv"
  ),
  row.names = FALSE
)
```